

```
exercise5.py x
3  def __init__(self, name, address):
4      self.name = name
5      self.address = address
6
7  def get_info(self):
8      return f"{self.name} lives at {self.address}."
9
10 class CardHolder:
11     def __init__(self, account_number):
12         self.account_number = account_number
13         self.balance = 0
14         self.credit_limit = 5000
15
16     def process_sale(self, price):
17         self.balance += price
18
19     def make_payment(self, amount):
20         self.balance -= amount
21
22 # declare child class here
23 class PlatinumClient(Person, CardHolder):
24     def __init__(self, name, address, account_number):
25         Person.__init__(self, name, address)
26         CardHolder.__init__(self, account_number)
27         self.cash_back = 0.02
28         self.rewards = 0
29
30     def process_sale(self, price):
31         super().process_sale(price)
32         self.rewards += price*self.cash_back
33
34 platinum = PlatinumClient("Sarah", "101 Main Street", 123364)
35 print(platinum.process_sale(100))
36 print(platinum.rewards)
37 print(platinum.balance)
38 print(platinum.make_payment(50))
39 print(platinum.balance)
```

Inheritance Exercise 5

Exercise 5

Use the parent classes to the left to help you solve this problem. The first parent is the `Person` class with some very generic information. The second class, `CardHolder`, is the class for a credit card holder. Create the child class `PlatinumClient`. This class inherits all of the attributes of both parent classes. In addition, the child class has the attributes `cash_back` and `rewards`. `cash_back` should be set to 0.02 and `rewards` should be set to `0`. Override the `process_sale` method so that 2% of each sale is added to `rewards`.

Expected Output

Declare an instance of the `PlatinumClient` class as shown below.

```
platinum = PlatinumClient("Sarah", "101 Main Street", 123364)
```

Your class should be able to execute the code below in the stated order and produce the given return values.

Order	Code	Return Value
1	<code>platinum.process_sale(100)</code>	N/A
2	<code>print(platinum.rewards)</code>	<code>2</code>
3	<code>print(platinum.balance)</code>	<code>100</code>
4	<code>platinum.make_payment(50)</code>	N/A
5	<code>print(platinum.balance)</code>	<code>50</code>
6	<code>print(platinum.get_info())</code>	<code>Sarah lives at 101 Main Street.</code>

TRY IT

None

2.0

100

None

50

Sarah lives at 101 Main Street.

Submit your code when ready

Here is one possible solution:

```
# parent classes
class Person:
    def __init__(self, name, address):
        self.name = name
        self.address = address

    def get_info(self):
        return f"{self.name} lives at {self.address}."

class CardHolder:
    def __init__(self, account_number):
        self.account_number = account_number
        self.balance = 0
        self.credit_limit = 5000

    def process_sale(self, price):
        self.balance += price

    def make_payment(self, amount):
        self.balance -= amount

# declare child class here
class PlatinumClient(Person, CardHolder):
    def __init__(self, name, address, account_number):
        Person.__init__(self, name, address)
        CardHolder.__init__(self, account_number)
        self.cash_back = 0.02
        self.rewards = 0

    def process_sale(self, price):
        self.balance += price
        self.rewards += self.cash_back * price
```

Check It!

LAST RUN on 12/28/2023, 1:18:47 AM

Test passed